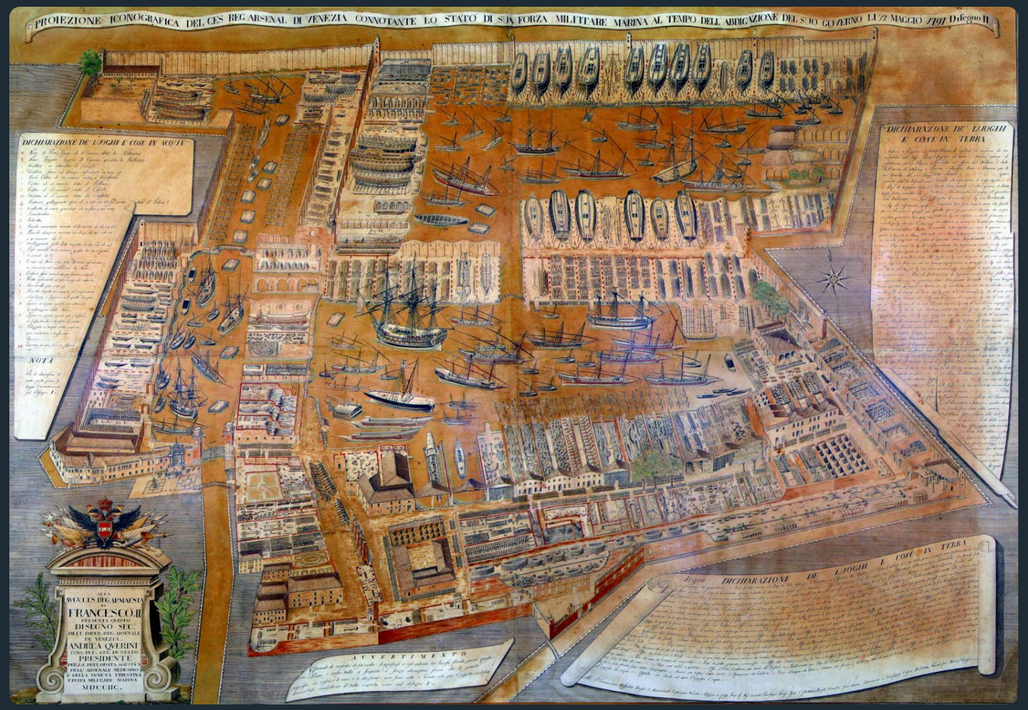
EXTENDED READING · W1.4

The Larger, the Weaker

Venice, Galileo and why a ship cannot simply be scaled up

1434	The Arsenal builds every ship from one length and fixed fractions
1638	Galileo asks why the great galleass needs more shoring, and answers with a beam
Today	Ship design meets the same limit whenever it scales up a reference ship



The Arsenal of Venice, Maffioletti, 1797,

The Arsenal of Venice, 1797



Centre of the basin: *La Vittoria*, a 70-gun ship of the line · Maffioletti, 1797

Course repository: this map ↗ · second scan ↗ (2,576 × 1,849 px)

The material flow, station by station: AllAboutLean.com ↗

A shipyard run like a factory:

Built in stages

Every component made on one site; a hull was towed along a canal and fitted out from the storehouses on each side.

Standard parts

Standard designs, assemblies and parts. Standardised, not yet interchangeable: large parts were matched to their hull.

Built to stock

Hulls and parts built in peacetime and stored under cover, then assembled into ships when war threatened.

A design office

A division for innovation and development: the map shows a models hall and a school of mathematics and drawing.

A Powerhouse of Specialised Work



One galley of Flanders: 41.2 m long, 6.1 m beam, ≈ 200 men

Michael of Rhodes, 1434, fol. 145b

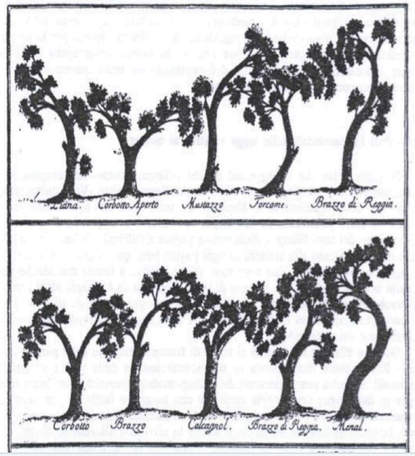
10 galleys
fitted out in six hours, 1436

100 galleys
ordered in reserve, 1545

≈ 750
carpenters, caulkers and oar makers, c. 1590

24–29 ha
of stores, workshops and slips

THE PRODUCTION LINE



Vittori, 1777: oaks grown into ship parts

1 Forest · oaks trained for 50–60 years into the curves of frames and knees

2 Templates · every piece cut to a *sesto*, a shaped template

3 Slip · hulls built and stored dry: at least 56 under cover in 1494

4 Canal · masts, ropes, oars and arms handed out from the windows

Its master shipwrights were kept **in state employment**, and their skills stayed in Venice.

3D virtual tour of the Arsenal ↗

Every Dimension Derived from One Length

Michael of Rhodes, 1434: a sailing ship

Dimension	Michael's rule	Metres	÷ keel
Keel	12 paces	20.9	1
Floor	12 − 3 = 9 **feet**	3.1	0.15
Ref. width	12 + 4 = 16 **feet**	5.6	0.27
Breadth	16 + 8 = 24 **feet**	8.3	**0.40**
Deck length	12 + 3 = 15 paces	26.1	**1.25**

L/B ≈ 3

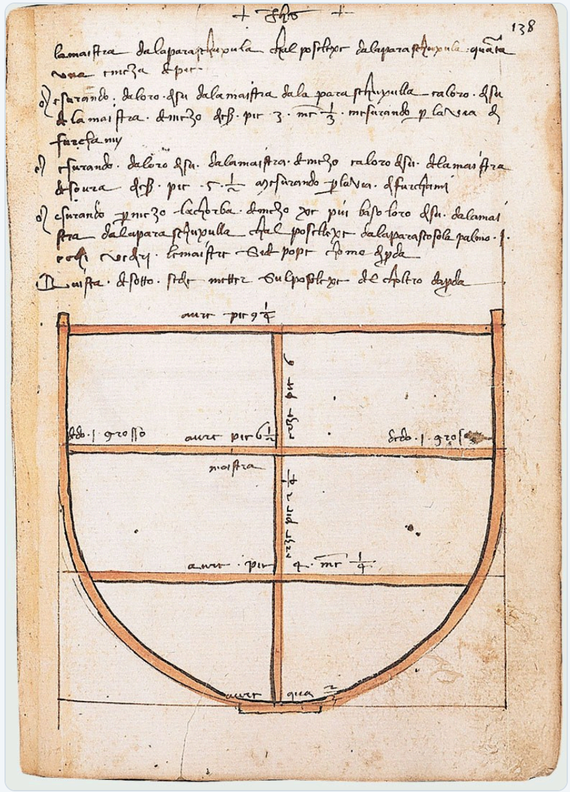
round ship: short and full, bulk cargo under sail

L/B ≈ 6.8

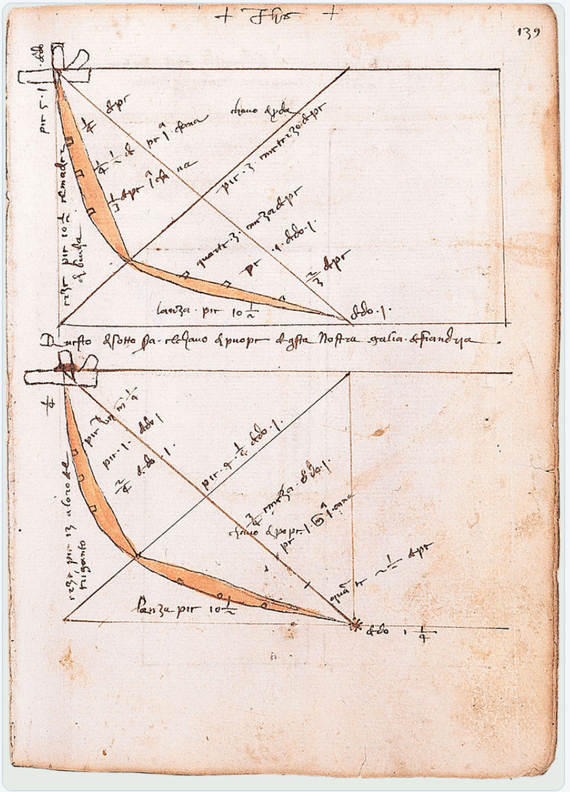
galley: long and slim, oars along the sides, speed

The trick: the keel is counted in **paces** and the answer read in **feet** (1 pace = 5 ft). So "keel − ¼" means floor = ¾ × keel ÷ 5 = **0.15 × keel**, for any keel.

> The ratio carries the mission: fix L/B and the ship type follows. No drawing of the whole hull was needed.

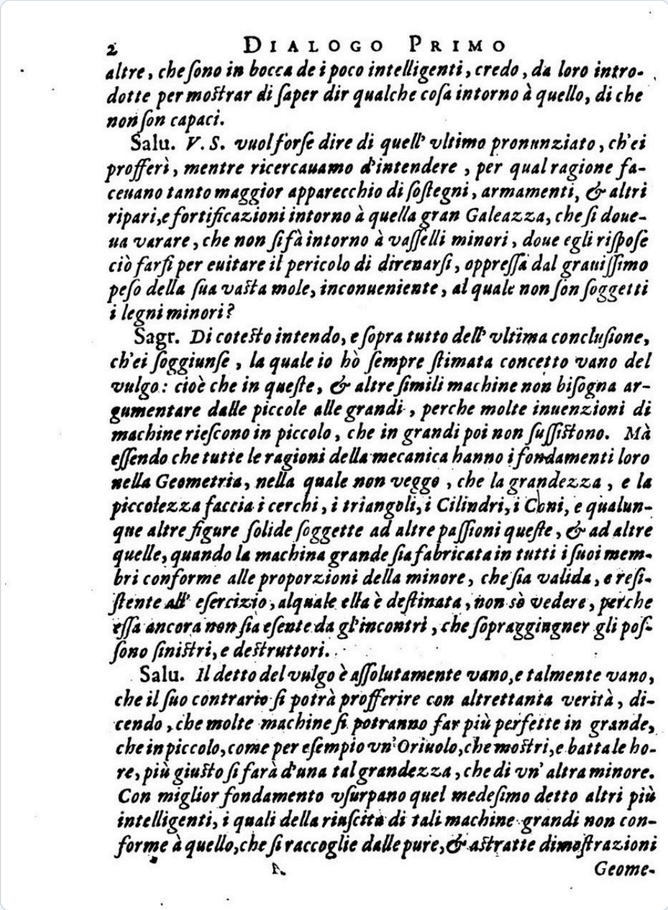


Fol. 138: a frame, its breadth fixed at set heights



Fol. 139b: stem and stern curves from a bisected right angle

The Question the Arsenal Put to Galileo



2 DIALOGO PRIMO

altre, che ſono in bocca de i poco intelligenti, credo, da loro introdotte per moſtrar di ſaper dir qualche coſa intorno à quello, di che non ſon capaci.

Salu. V. S. vuol forſe dire di quell' vltimo pronunziato, ch'ei profferì, mentre ricercauamo d'intendere, per qual ragione faceuano tanto maggior apparecchio di ſoſtegni, armamenti, & altri ripari, e fortificazioni intorno à quella gran Galeazza, che ſi doueua varare, che non ſi fà intorno à vaſſelli minori, doue egli riſpoſe ciò farſi per euitare il pericolo di direnarſi, oppreſſa dal graui ſſimo peſo della ſua vaſta mole, inconueniente, al quale non ſon ſoggetti i legni minori?

Sagr. Di coteſto intendo, e ſopra tutto dell' vltima concluſione, ch'ei ſoggiunſe, la quale io hò ſempre ſtimata concetto vano del vulgo: cioè che in queſte, & altre ſimili machine non biſogna argumentare dalle piccole alle grandi, perche molte inuenzioni di machine rieſcono in piccolo, che in grandi poi non ſuſſiſtono. Mà eſſendo che tutte le ragioni della mecanica hanno i fondamenti loro nella Geometria, nella quale non veggo, che la grandezza, e la piccolezza faccia i cerchi, i triangoli, i Cilindri, i Coni, e qualunque altre figure ſolide ſoggette ad altre paſſioni queſte, & ad altre quelle, quando la machina grande ſia fabricata in tutti i ſuoi membri conforme alle proporzioni della minore, che ſia valida, e reſiſtente all' eſercizio, alquale ella è deſtinata, non sò vedere, perche eſſa ancora non ſia eſente da gl'incontri, che ſopraggiugner gli poſſono ſiniſtri, e deſtruttori.

Salu. Il detto del vulgo è aſſolutamente vano, e talmente vano, che il ſuo contrario ſi potrà profferire con altrettanta verità, dicendo, che molte machine ſi potranno far più perfette in grande, che in piccolo, come per eſempio vn' Oriuolo, che moſtri, e batta le hore, più giuſto ſi farà d'una tal grandezza, che di vn' altra minore. Con miglior fondamento vſurpano quel medeſimo detto altri più intelligenti, i quali della rinſcita di tali machine grandi non conforme à quello, che ſi raccoglie dalle pure, & aſtratte dimoſtrazioni

A Geome-

Galilei, *Discorsi*, Leida, 1638, p. 2

1638 · THE DIALOGUE

Why does the great galleass need far more shoring than a small vessel?

> *“…per evitare il pericolo di direnarsi, oppressa dal gravissimo peso della sua vasta mole”*
>
> The old master: so that she does not break under her own great weight.

> *“quanto più sarà grande, tanto a proporzione sarà più debole”*
>
> Salviati: the larger, the weaker in proportion, even in the same material and shape.

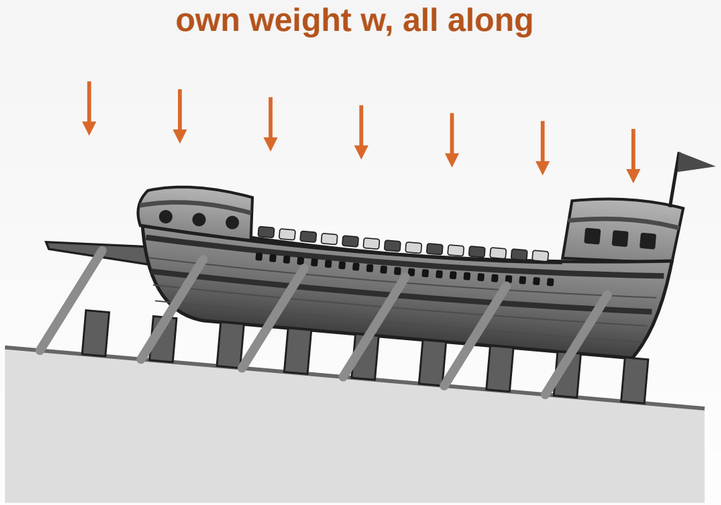
1593 · THE RECORD

Behind the dialogue, a real consultation: the Arsenal asked Galileo about the oars of its large galleys (letters of 22 and 28 March 1593).

From the Galleass to Galileo’s Beam

1 · On the slip

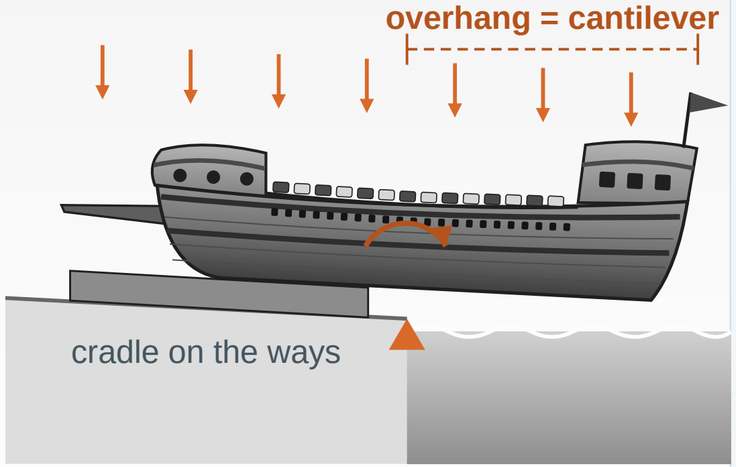
many shores, short spans



keel blocks and shores (sostegni)
each span short: small bending

2 · Launch

the stern runs past the end of the ways



peak bending at the end of the ways:
the hull may break its back (direnarsi)

3 · Galileo’s reduction

one beam, a weight E at its end


114 DIALOGO SECONDO

fin quì dichiarate, non sarà difficile l'intender la ragione, onde auuenga, che vn Prisma, ò Cilindro solido di vetro, acciaio, legno, ò altra materia frangibile, che sostenuto per lungo sotterrà gravissimo peso, che gli sia attaccato, mà in trauerso (come poco fà dicemmo) da minor peso assai potrà tal volta essere spezzato, secondo che la sua lunghezza eccederà la sua grossezza. Imperò che figuriamoci il Prisma solido AB, CD fitto in vn muro dalla parte AB, e nell'altra estremità s'intenda la forza del Peso E, (intendendo sempre il muro esser eretto all'Orizonte, & il Prisma, ò Cilindro fitto nel muro ad angoli retti) è manifesto che douendosi spezzare si romperà nel luogo B, doue il taglio del muro serue per sostegno, e la BC per la parte della Leua, doue si pone la forza, e la grossezza del solido BA, e l'altra parte della Leua, nella quale è posta la resistenza, che consiste nello staccamento, che s'hà da fare della parte del solido BD, che è fuor del muro, da quella che è dentro; e per le cose dichiarate il momento della forza posta in C al momento della resistenza che stà nella

Prop. I.

Galilei, *Discorsi*, 1638, Day Two, Prop. I, p. 114

> Galileo keeps only what decides the answer: **one beam, its own weight, a load at the free end**.

A Scaling Problem Everyone Knows



“*un osso allungato solamente tre volte*”: a bone lengthened only three times. Galilei, 1638, p. 129

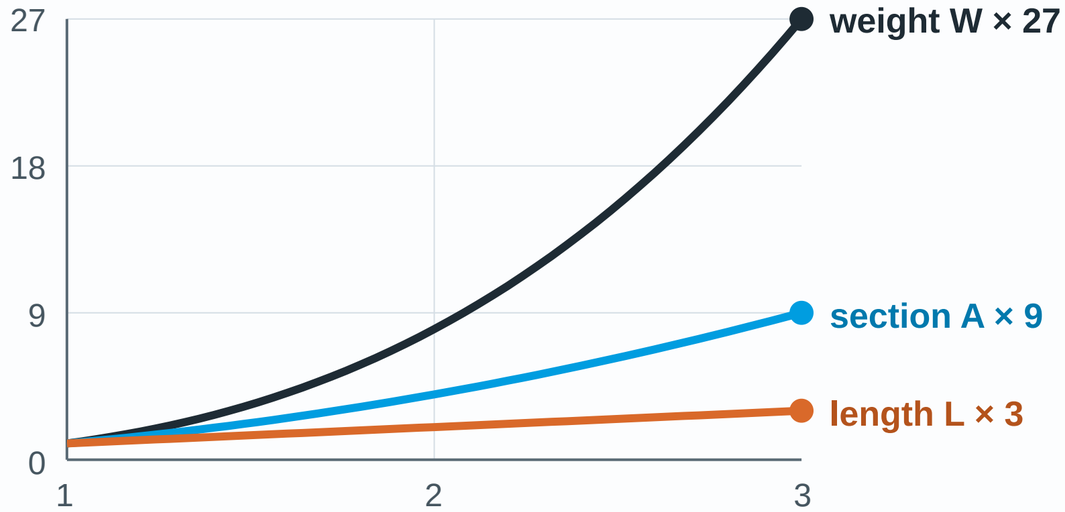
“impossibil sarebbe fabbricar navilii, palazzi o templi vastissimi”
It would be impossible to build ships, palaces or temples of immense size.

“…some which appear feasible in models, but when they have begun to increase in size are impracticable”
Vitruvius, *De architectura* X.16.5, 1st c. BC

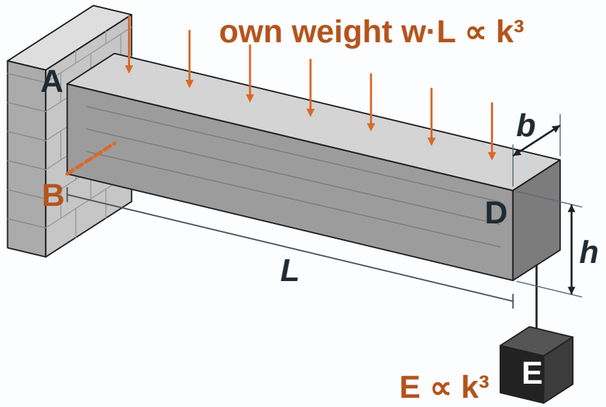
Builders had met the effect since antiquity, but **no one had stated the law or explained it before Galileo, 1638**.

The square–cube law: make any body three times larger

Quantity, all lengths × k	grows as	k = 3	k = 10
Length L	k	3	10
Section A	k^2	9	100
Volume, mass, weight W	k^3	27	1,000



The First Structural Optimisation Problem



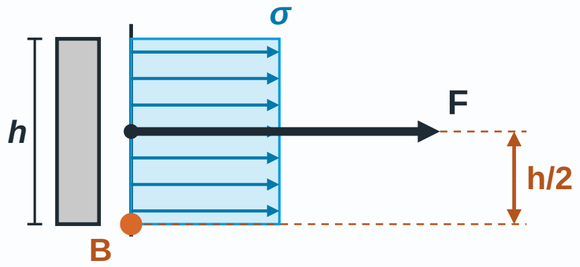
Symbols used here and on slide 8

L, b, h	length, breadth, depth	I	second moment of area, b·h³/12
k	scale factor	y	distance to the neutral axis
w	weight per metre, ρ·g·b·h	Z	section modulus, I/y = b·h²/6
ρ	density, steel 7,850 kg/m³	E	weight hung at the free end, N
M	bending moment, N·m	g	gravity, 9.81 m/s²
σ	stress, MPa = N/mm²		

Below · **Grey:** end face, depth h · **Blue arrows:** pull of each fibre, σ · **F:** all pulls added up

Galileo, 1638

all fibres pull equally; hinge at the bottom edge

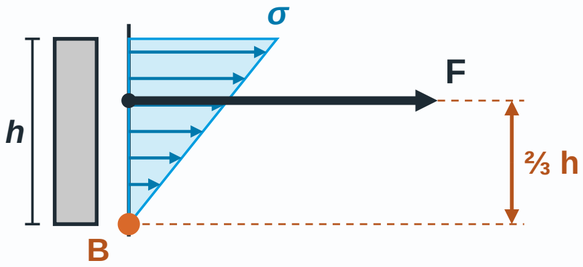


F = σ·b·h, arm = h/2

M = F·arm = ½ σ b h²

Mariotte, 1686

pull grows from zero at the bottom edge

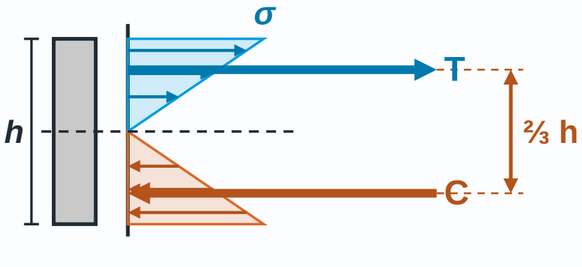


F = ½ σ·b·h, arm = ⅔ h

M = F·arm = ⅓ σ b h²

Parent 1713 · Navier 1826

top pulls, bottom pushes, middle at rest

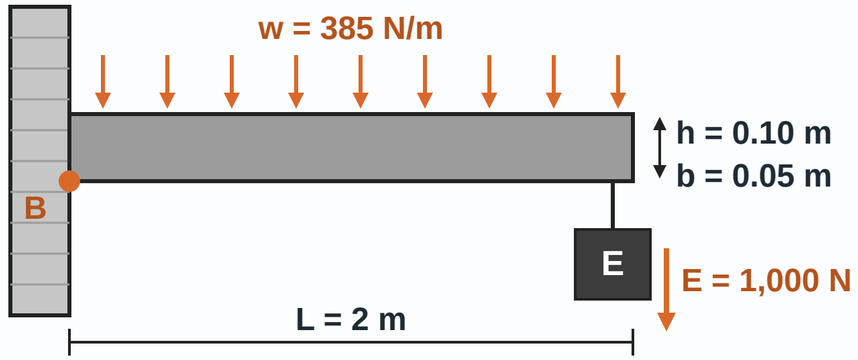


T = C = ¼ σ·b·h, arm = ⅔ h

M = T·arm = ⅙ σ b h²

Worked Example: Own Weight plus the Weight E

The model, k = 1: a steel beam



1 · Own weight

w = ρ·g·b·h = 7,850 × 9.81 × 0.05 × 0.10 = 385 N/m

2 · Moment at the wall: Archimedes' lever

M = E·L + w·L²/2 = 1,000 × 2 + 385 × 2²/2 = 2,770 N·m

3 · Section

b·h² = 0.05 × 0.10² = 0.0005 m³

4 · Stress, Navier

σ = M / (⅙·b·h²) = 2,770 / 0.0000833 = 33.2 MPa

Quantity	Model, k = 1	Copy, k = 10	grows
Size L × b × h	2 × 0.05 × 0.10 m	20 × 0.5 × 1.0 m	k
Own weight per metre w	385 N/m	38,500 N/m	k²
Beam weight w·L	770 N	770,000 N	k³
Weight at the end E	1,000 N	1,000,000 N	k³
Moment M	2,770 N·m	27.7 million N·m	k⁴
Section b·h²	0.0005 m³	0.5 m³	k³
σ Galileo, c = ½	11.1 MPa	111 MPa	k
σ Mariotte, c = ⅓	16.6 MPa	166 MPa	k
σ Navier, c = ⅙	33.2 MPa	332 MPa	k
Depth h to reach yield	0.034 m	1.23 m (b = 0.5 m)	–

How deep must the copy be? (k = 10, b = 0.5 m)

w = ρ·g·b·h = 7,850 × 9.81 × 0.5 × h = 38,500·h N/m

M = E·L + w·L²/2 = 20.0 + 7.70·h million N·m

σ = M / (b·h²/6) = 12·M / h²

to reach yield, σ = 235 MPa → **h = 1.23 m**

to keep the model's stress, σ = 33.2 MPa → **h = 4.42 m**

How Must the Section Grow?

Slide 6: copy every length by k

mass and load ∝ **k³**, section ∝ **k²** → stress ∝ **k**

The section grows one power of k too slowly.

Slide 8: when does the copy fail?

Plain copy of the 2 m beam: σ = 33.2 MPa × k

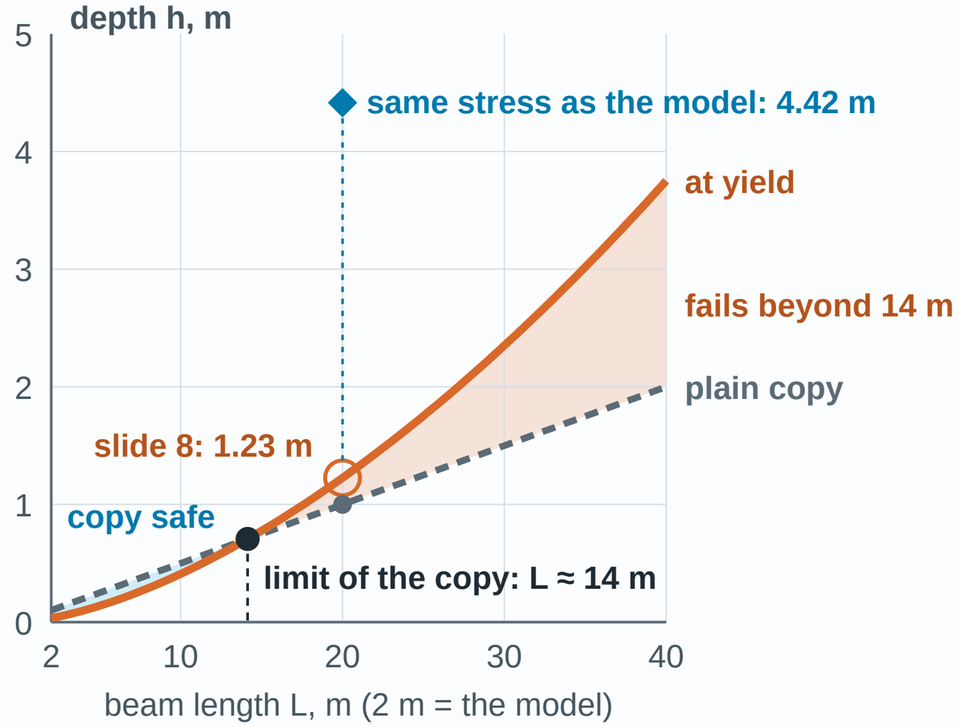
It reaches the 235 MPa yield at **k = 235/33.2 ≈ 7, L ≈ 14 m**

Longer than that, the same proportions fail.

Depth needed to reach yield (copy's breadth kept)

L = 20 m: **1.23 m** (plain copy 1.0 m)

L = 40 m: **3.75 m** (plain copy 2.0 m)



Depth needed to just reach yield (235 MPa), against a plain copy of the 2 m model; blue: the depth that keeps the model's stress at 20 m (slide 8)

Every copy has a size limit, here **14 m**; beyond it the section must grow faster than the size. **Proportions cannot simply be copied.**

Proportions Hold Only Within a Range of Size

1 · SLIDES 1–3

Venice built by proportion

Every dimension a fixed fraction of the keel. It worked because each new ship was close in size to the last.

2 · SLIDES 4–6

A copy is weaker

Scale every length by k: weight grows as k^3, the section as k^2, so the stress grows as **k**.

3 · SLIDE 7

The law outlived the theory

Galileo's beam constant was 3 times off, ½ against Navier's ⅙, but stress ∝ k holds in every theory.

4 · SLIDES 8–9

Every copy has a limit

The 2 m steel beam is safe; its copy yields beyond 14 m. At 20 m it needs 1.23 m of depth, not 1.0 m.

What "non-linear" means

A quantity is linear in size when doubling the size doubles it: $y = a \cdot x$, a straight line through the origin. Mass is not: $y = a \cdot x^n$ with $n \neq 1$. A copy's mass grows as k^3, and the beam of slide 9, kept at yield, grows faster still: from 20 m to 40 m its mass rises **12 times**, not 2 (linear) or 8 (copy).

For ship design: ship mass varies non-linearly with size

Weights do not follow the main dimensions in proportion, so ratios taken from similar ships (Lecture 1.4) hold only inside their size range. Beyond it, the size must come from physics, $\Delta = \rho \cdot C_B \cdot L \cdot B \cdot T$ and $\sigma = M/Z$, and the weights from power laws with exponents other than 1 (Lecture 1.5).